SCSI 1113: COMPUTATIONAL MATHEMATICS

20172018-2

TUTORIAL 1.2

Due Date: 16 March 2018

1. For $u = \begin{bmatrix} 4 \\ 7 \\ -3 \\ 2 \end{bmatrix}$ and $v = \begin{bmatrix} 5 \\ -2 \\ 8 \\ 1 \end{bmatrix}$, find

   (a) $6(2u - v)$ (b) $\|u + v\|$ (c) $u \cdot v$

   (d) show that $\|u + v\| \leq \|u\| + \|v\|$

   (e) the distance for u and v in the Euclidean space $\mathcal{R}^4$

   (f) the norm of v

2. Show that $u = \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$ can be expressed as a linear combination of $u_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}, u_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

3. Determine whether the vectors given form a linearly dependent set or linearly independent set.

   (a) $u_1 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix}, u_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$

   (b) $v_1 = \begin{bmatrix} 3 \\ -1 \\ -1 \\ 2 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 3 \\ -1 \\ 0 \\ 1 \end{bmatrix}$